18. The mass of one mole of carbon

19. No, of moles of oxygen = 8 g/32 g
= 0.25 moles of oxygen

No. of moles of sulphur dioxide = 8 g / 64 g
= 0.125 moles of sulphur dioxide

Ratio between the no. of molecules = 0.25: 0.125
= 2:1

20. $AgNO_3 + KCl \rightarrow KNO_3 + AgCl$

50 mL of 8.5 % solution contains 4.25 g of $AgNO_3$

No. of moles of AgNO3 present in 50 mL of 8.5 % $AgNO_3$ solution

= Mass / Molar mass

= 4.25 / 170

= 0.025 moles

Similarly, No of moles of KCl present in 100 mL of 1.865 % KCl solution

= 1.865 / 74.5

= 0.025 moles

So total amount of AgCl formed is 0.025 moles (based on the stoichiometry)

Amount of AgCl present in 0.025 moles of AgCl

= no. of moles x molar mass

= **0.025 x 143.5 = 3.59 g**

21. No. of moles of a gas that occupies a volume of 612.5 mL at room temperature and pressure (25 ^{0}C and 1 atm pressure)

= 612.5×10^{-3} L / 24.5 $Lmol^{-1}$

= 0.025 moles

We know that,

Molar mass = Mass / no. of moles

= **1.1 g /0.025 mol = 44 g mol^{-1}**

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